



NETWORKING GUIDE

دليل الشبكات الشامل

شرح كامل للشبكات من الأساسيات للاحتتراف
مع الأوامر والأدوات العملية

OSI Model | Protocols | Subnetting | Routing | Security | Analysis

OSI MODEL - 7 LAYERS

الطبقات السبع OSI نموذج

OVERVIEW / نظرة عامة

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes network communication into 7 layers. يحدد اتصالات الشبكة في 7 طبقات، من الكيبل المادي حتى التطبيق OSI نموذج.

LAYER	NAME	PROTOCOLS	DEVICES	UNITS	الوصف
7	Application	HTTP, FTP, SMTP, DNS, SSH, POP3, IMAP, DHCP	Gateway, Browser, Server	Data	طبقة التطبيق - واجهة المستخدم والبرامج
6	Presentation	SSL/TLS, JPEG, GIF, MPEG, ASCII	Gateway	Data	طبقة العرض - تشفير وضغط البيانات
5	Session	NetBIOS, RPC, PPTP, SIP	Gateway	Data	طبقة الجلسة - إدارة الاتصالات
4	Transport	TCP, UDP, SCTP, DCCP	Firewall, Load Balancer	Segment	طبقة النقل - نقل موثوق/غير موثوق
3	Network	IP, ICMP, ARP, OSPF, BGP, RIP	Router, L3 Switch	Packet	طبقة الشبكة - التوجيه والعناوين
2	Data Link	Ethernet, PPP, MAC, VLAN, ARP	Switch, Bridge, NIC	Frame	طبقة ربط البيانات - عناوين MAC
1	Physical	10BASE-T, 100BASE-T, Fiber, DSL	Hub, Repeater, Cable	Bit	طبقة مادية - كيابل وإشارات

TCP/IP MODEL

TCP/IP نموذج

LAYER	PROTOCOLS	OSI EQUIVALENT	الوظيفة
Application	HTTP, FTP, SMTP, DNS, SSH, Telnet	Layers 5-7	التطبيقات والخدمات
Transport	TCP, UDP	Layer 4	نقل البيانات
Internet	IP, ICMP, ARP, IGMP	Layer 3	العنونة والتوجيه
Network Access	Ethernet, Wi-Fi, PPP	Layers 1-2	الوصول المادي للشبكة

KEY NETWORK CONCEPTS

مفاهيم أساسية

TCP VS UDP / مقارنة

FEATURE	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed delivery	No guarantee
Ordering	In-order delivery	No ordering
Speed	Slower (ACK overhead)	Faster
Port range	1-65535	1-65535
Header size	20-60 bytes	8 bytes
Use cases	Web (80), Email (25), SSH (22)	DNS (53), DHCP (67), VoIP

PORT RANGES / نطاقات المنافذ

Well-known: 0-1023 (HTTP 80, HTTPS 443, SSH 22, FTP 21, DNS 53) | Registered: 1024-49151 | Dynamic: 49152-65535

\$ NETWORKING - IP ADDRESSING & SUBNETTING

IPV4 ADDRESSING

IPv4 العنونة

STRUCTURE / التركيب

IPv4: 32-bit - 4 octets - e.g., 192.168.1.1 (0-255 each octet) | Binary: 11000000.10101000.00000001.00000001
يتكون من 32 بت مقسمة إلى 4 أقسام، كل قسم 8 بت (0-255) IPv4 عنوان

CLASS	RANGE	DEFAULT MASK	CIDR	HOSTS	USAGE
A	1.0.0.0 - 126.255.255.255	255.0.0.0	/8	16M	Large networks
B	128.0.0.0 - 191.255.255.255	255.255.0.0	/16	65K	Medium networks
C	192.0.0.0 - 223.255.255.255	255.255.255.0	/24	254	Small networks
D	224.0.0.0 - 239.255.255.255	N/A	N/A	N/A	Multicast
E	240.0.0.0 - 255.255.255.255	N/A	N/A	N/A	Reserved

PRIVATE IP RANGES (RFC 1918) / عناوين خاصة

Class A: 10.0.0.0/8 (10.0.0.0 - 10.255.255.255) | Class B: 172.16.0.0/12 (172.16.0.0 - 172.31.255.255) | Class C: 192.168.0.0/16 (192.168.0.0 - 192.168.255.255) | Loopback: 127.0.0.0/8

SUBNETTING & CIDR

CIDR تقسيم الشبكات

CIDR NOTATION / ترميز CIDR

Format: IP/prefix (e.g., 192.168.1.0/24) | Prefix = number of network bits | /24 = 255.255.255.0 = 256 addresses (254 usable)

CIDR	MASK	SUBNETS	HOSTS	CIDR	MASK	SUBNETS	HOSTS
/30	255.255.255.252	64	2	/22	255.255.252.0	64	1022
/29	255.255.255.248	32	6	/21	255.255.248.0	32	2046
/28	255.255.255.240	16	14	/20	255.255.240.0	16	4094
/27	255.255.255.224	8	30	/19	255.255.224.0	8	8190
/26	255.255.255.192	4	62	/18	255.255.192.0	4	16382
/25	255.255.255.128	2	126	/17	255.255.128.0	2	32766
/24	255.255.255.0	1	254	/16	255.255.0.0	1	65534

SUBNETTING FORMULA / معادلة التقسيم

Subnets = 2^(borrowed bits) | Hosts per subnet = 2^(remaining bits) - 2 | e.g., /26 from /24: borrow 2 bits = 4 subnets, 2^6-2 = 62 hosts each

IPV6 BASICS

IPv6 أساسيات

STRUCTURE / التركيب

IPv6: 128-bit - 8 groups of 4 hex digits - e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Shortening: Remove leading zeros, :: replaces consecutive zero groups

Types: Unicast, Multicast, Anycast | No broadcast, no NAT needed

TYPE	PREFIX	DESCRIPTION	الوصف
Global Unicast	2000::/3	Internet routable	عناوين عامة للإنترنت
Unique Local	FC00::/7	Private (like 10.0.0.0/8)	عناوين خاصة
Link-Local	FE80::/10	Local segment only	محلية للقطاع فقط
Multicast	FF00::/8	One-to-many (replaces broadcast)	بث جماعي
Loopback	::1/128	Localhost	المحلي

\$ NETWORKING - PROTOCOLS DEEP DIVE

APPLICATION LAYER PROTOCOLS

وتوكولات طبقة التطبيق

PROTOCOL	PORT	COMMAND	DESCRIPTION	الوصف
HTTP	80	curl http://ex.com	HyperText Transfer Protocol	النص التشعبي
HTTPS	443	curl https://ex.com -k	HTTP over SSL/TLS	HTTP مشفر
FTP	21	ftp ftp://ex.com	File Transfer Protocol	الملفات
SFTP	22	sftp user@host	SSH File Transfer	ملفات آمن
SSH	22	ssh user@host	Secure Shell	م آمن عن بعد
Telnet	23	telnet host port	Remote login (unencrypted)	م عن بعد غير مشفر
SMTP	25	telnet host 25	Simple Mail Transfer	البريد
POP3	110	telnet host 110	Post Office Protocol v3	قبيل البريد
IMAP	143	telnet host 143	Internet Message Access	البريد
DNS	53	dig example.com	Domain Name System	الاسماء
DHCP	67/68	dhclient eth0	Dynamic Host Configuration	عناوين
SNMP	161	snmpwalk -v2c -c public host	Network Management	الشبكة
NTP	123	ntpdate pool.ntp.org	Network Time Protocol	زمن الوقت
LDAP	389	ldapsearch -x -h host	Lightweight Directory Access	المستخدمين
SMB	445	smbclient -L host	Server Message Block	مشاركة الملفات
RDP	3389	xfreerdp /v:host	Remote Desktop Protocol	مكتب عن بعد
	3306	mysql -h host -u root -p		

MySQL			MySQL Database	MySQL بيانات
PostgreSQL	5432	psql -h host -U postgres	PostgreSQL Database	PostgreSQL بيانات
Redis	6379	redis-cli -h host	Redis Cache	Redis تخزين مؤقت
MongoDB	27017	mongo --host host	MongoDB Database	MongoDB بيانات
MQTT	1883	mosquitto_sub -h host -t topic	MQ Telemetry Transport	MQTT اتصالات الأشياء

ROUTING PROTOCOLS

بروتوكولات التوجيه

Protocol	Type	Metric	AD Value	Port/Protocol	Description
RIP	Distance Vector	Hop count (max 15)	120	UDP 520	Routing Information Protocol - simple, small networks
OSPF	Link State	Cost (bandwidth)	110	IP 89	Open Shortest Path First - enterprise, fast convergence
EIGRP	Hybrid /Adv Distance Vector	Composite (BW+delay)	90 (internal)/170 (ext)	IP 88	Cisco proprietary, advanced metrics
BGP	Path Vector	Path attributes	20 (eBGP)/200 (iBGP)	TCP 179	Border Gateway Protocol - internet routing
IS-IS	Link State	Metric (default 10)	115	CLNS	Intermediate System - ISP backbone

AD (Administrative Distance) / المسافة الإدارية
Connected: 0 Static: 1 eBGP: 20 EIGRP: 90 OSPF: 110 IS-IS: 115 RIP: 120 iBGP: 200 Unknown: 255

\$ NETWORKING - PRACTICAL COMMANDS

ESSENTIAL NETWORK COMMANDS

الشبكات الأساسية

Command	Example	Description	الوصف
ping	ping -c 4 google.com	Basic connectivity test	مصال الأساسية
ping -f google.com (flood ping)			
tracert	tracert google.com	Trace packet path	الحزم
tracert -n google.com (no DNS)			
tracertpath	tracertpath google.com	Trace without root	صلاحيات
mtr	mtr -r google.com	Continuous tracert + ping	بر مع ping
mtr -c 10 google.com			
ip	ip addr show	Show IP addresses	اوين
ip route show (routing table)			
ip link set eth0 up/down			
ip neigh (ARP cache)			
ifconfig	ifconfig eth0	Interface configuration	جهات
ifconfig eth0 192.168.1.100 netmask 255.255.255.0			
iwconfig	iwconfig wlan0	Wireless config	اللكي
route	route -n	Show routing table	جيه
route add default gw 192.168.1.1			
arp	arp -a	ARP cache display	المخبة
arp -d 192.168.1.1 (delete entry)			
netstat	netstat -tulpn	Listening ports	ستمعة
netstat -an (all connections)			
netstat -r (routing table)			
ss	ss -tulpn	Socket statistics	المقابس
ss -s (summary)			
nmap	nmap -sn 192.168.1.0/24	Network discovery	شبكة
nmap -sV -p 1-1000 target			
tcpdump	tcpdump -i eth0	Packet capture	م
tcpdump port 80 -w capture.pcap			
dig	dig google.com ANY	DNS lookup	DN
dig -x 8.8.8.8 (reverse)			
nslookup	nslookup google.com	DNS query	DN
host	host -a google.com	DNS lookup	
whois	whois google.com	Domain lookup	ق
curl	curl -I https://ex.com	HTTP headers	HTTP
curl -v https://ex.com (verbose)			

wget	wget https://ex.com/file	File download	تنزيل الملفات
nc (netcat)	nc -lvnp 4444	TCP listener/connector	مستمع/مستفيد TCP
echo 'test' nc IP 80			
ncat	ncat -lvnp 4444 --ssl	SSL-enabled netcat	مع تشفير
socat	socat TCP-L:4444,fork EXEC:/bin/bash	Bidirectional relay	الاتجاه
ssh	ssh -i key.pem user@host	Secure remote shell	عن بعد
ssh -L 8080:localhost:80 user@host (tunnel)			
scp	scp file.txt user@host:/tmp/	Secure copy	
rsync	rsync -avz dir/ user@host:dir/	File sync	ات
iptables	iptables -L -v -n	Firewall rules	ار النار
iptables -A INPUT -p tcp --dport 22 -j ACCEPT			
ufw	ufw status verbose	Uncomplicated firewall	مبسط
ufw allow 80/tcp			
dhclient	dhclient eth0	DHCP client	DH
hostnamectl	hostnamectl set-hostname newname	Set hostname	الجهاز
resolvectl	resolvectl status	DNS resolver status	DNS
systemd-resolve	systemd-resolve --status	Systemd resolver	system

WIRELESS NETWORKING COMMANDS

لشبكات اللاسلكية

COMMAND	EXAMPLE	DESCRIPTION	الوصف
iwconfig	iwconfig wlan0	Wireless interface info	الواجهة اللاسلكية
iwconfig wlan0 mode monitor			
iwlist	iwlist wlan0 scan	Scan Wi-Fi networks	كشكات اللاسلكية
iwlist wlan0 channel			
nmcli	nmcli dev wifi list	NetworkManager CLI	الشبكة
nmcli dev wifi connect SSID password PASSWORD			
nmcli	nmcli connection show	List connections	مصالات
rfkill	rfkill list	Radio kill switch	ف الراديو
rfkill unblock wifi			
airmon-ng	airmon-ng start wlan0	Monitor mode	مقبة
airodump-ng	airodump-ng wlan0mon	Packet capture	مزم اللاسلكية
iw dev	iw dev wlan0 link	Connection status	ال
wpa_supplicant	wpa_supplicant -i wlan0 -c config.conf	WPA authentication	WP
dhcpcd	dhcpcd wlan0	DHCP for wireless	سلكي

\$ NETWORKING - SECURITY

FIREWALL CONCEPTS

جدار الناري

FIREWALL TYPES / أنواع الجدران النارية

1) Packet Filter (Layer 3/4) - iptables/nftables | 2) Stateful (tracks connections) - firewallld/ufw | 3) Application (WAF/Proxy) - WAF/Proxy | 4) Next-Gen (NGFW) - Deep packet inspection, IPS, app awareness

TOOL	EXAMPLE	DESCRIPTION	الوصف
iptables	iptables -P INPUT DROP	Default deny incoming	تراضياً
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT			
iptables -A INPUT -p tcp --dport 22 -j ACCEPT			
iptables -A INPUT -s 10.0.0.0/8 -j DROP			
iptables -A INPUT -j LOG --log-prefix 'DROPPED: '			
iptables-save > /etc/iptables/rules.v4			
nftables	nft add rule inet filter input tcp dport 22 accept	Modern firewall	
nft list ruleset			
ufw	ufw default deny incoming && ufw default allow outgoing	Policy setup	
ufw allow 22/tcp && ufw allow 80/tcp && ufw allow 443/tcp			
ufw deny from 10.0.0.0/8			
ufw enable && ufw status verbose			
firewalld	firewall-cmd --zone=public --add-service=http --permanent	Zone-based firewall	لق
firewall-cmd --zone=internal --add-source=192.168.1.0/24			
firewall-cmd --runtime-to-permanent && firewall-cmd --reload			

TOOL	EXAMPLE	DESCRIPTION	الوصف
snort	snort -i eth0 -c snort.conf -A console	Network IDS/IPS	مراقبة الشبكة
	snort -r capture.pcap -c snort.conf		
suricata	suricata -i eth0 -c suricata.yaml	High-performance IDS	مراقبة الشبكة
	suricata -r capture.pcap -c suricata.yaml		
zeek	zeek -i eth0	Network security monitor	مراقبة الشبكة
	zeek -r capture.pcap		
fail2ban	fail2ban-client status sshd	Brute-force protection	الحماية من هجمات القوة الغاشقة
	fail2ban-client set sshd maxretry 3		
ossec	ossec-control start	Host-based IDS	مراقبة النظام
	ossec-control status		
wazuh	wazuh-control start	Enterprise SIEM	مراقبة النظام
lynis	lynis audit system	Security audit	مراجعة الأمان
rkhunter	rkhunter --check --skip-keypress	Rootkit detection	كشف الرقعة
chkrootkit	chkrootkit	Rootkit detector	كشف الرقعة
aide	aide --check	File integrity	مراقبة سلامة الملفات
clamav	clamscan -r /home	Antivirus	مكافحة الفيروسات

PROTOCOL/TOOL	EXAMPLE	DESCRIPTION	الوصف
OpenVPN	openvpn --config client.ovpn	SSL/TLS VPN	VPN
	openvpn --genkey --secret static.key (static key)		
WireGuard	wg-quick up wg0	Modern fast VPN (UDP)	VPN
	wg genkey tee privatekey wg pubkey > publickey		
IPsec	ipsec start && ipsec status	IPsec VPN (IKEv1/v2)	VPN
	strongswan start		
L2TP	xl2tpd -c /etc/xl2tpd/xl2tpd.conf	L2TP VPN	L2TP
PPTP	pptpsetup --create VPN --server host --user user --pass pass	PPTP VPN (legacy)	PPTP
SSH Tunnel	ssh -D 1080 user@host	SOCKS proxy via SSH	SSH
	ssh -L 8080:localhost:80 user@host		
SSH -R	ssh -R 8080:localhost:80 user@host	Reverse tunnel	SSH

TOOL	EXAMPLE	DESCRIPTION	الوصف
tcpdump	tcpdump -i eth0 -nn -X	CLI packet analyzer	التتبع
	tcpdump -i eth0 port 53 (DNS traffic)		
	tcpdump -i eth0 host 192.168.1.1		
	tcpdump -i eth0 net 192.168.1.0/24		
	tcpdump -w capture.pcap -i eth0 -s 0		
wireshark	wireshark	GUI packet analyzer	التتبع
	wireshark -k -i eth0		
tshark	tshark -i eth0	CLI Wireshark	التتبع
	tshark -Y "http.request" -r capture.pcap		
	tshark -T fields -e ip.src -e ip.dst -e http.request.uri		
ettercap	ettercap -T -M arp:remote // //	MiTM sniffing	التتبع
	ettercap -G (GUI mode)		
bettercap	bettercap -eval "net.sniff on; arp.spoof on"	Modern MiTM	التتبع
	bettercap -iface eth0		
driftnet	driftnet -i eth0	Image sniffing	التتبع
tcpflow	tcpflow -i eth0 port 80	TCP stream reassembly	التتبع
chaosreader	chaosreader capture.pcap	Session reconstruction	التتبع
ngrep	ngrep -i "password" port 80	Network grep	التتبع
	ngrep -d eth0 -i "user" tcp port 80		

SERVICE	CONFIG FILE	COMMANDS	الوصف
DHCP	/etc/dhcp/dhcpd.conf	systemctl start isc-dhcp-server	
DNS	/etc/bind/named.conf	systemctl start bind9	
Apache	/etc/apache2/sites-available/000-default.conf	systemctl start apache2	
Nginx	/etc/nginx/nginx.conf	systemctl start nginx	
FTP	/etc/vsftpd.conf	systemctl start vsftpd	
Samba	/etc/samba/smb.conf	systemctl start smbd	
SSH	/etc/ssh/sshd_config	systemctl start sshd	
SNMP	/etc/snmp/snmpd.conf	systemctl start snmpd	
NTP	/etc/ntp.conf	systemctl start ntp	
Postfix	/etc/postfix/main.cf	systemctl start postfix	
OpenVPN	/etc/openvpn/server.conf	systemctl start openvpn@server	
WireGuard	/etc/wireguard/wg0.conf	wg-quick up wg0	

BANDWIDTH & PERFORMANCE MONITORING

TOOL	EXAMPLE	DESCRIPTION	الوصف
nload	nload eth0	Real-time bandwidth	
iftop	iftop -i eth0	Bandwidth by host	س
nethogs	nethogs eth0	Bandwidth by process	
iptraf-ng	iptraf-ng	Network statistics	
bmon	bmon -p eth0	Bandwidth monitor	
vnstat	vnstat -i eth0 -l	Traffic logger	
vnstat -d (daily) vnstat -m (monthly)			
darkstat	darkstat -i eth0	Web traffic monitor	
speedtest-cli	speedtest-cli	Speed test	
iperf3	iperf3 -s (server) / iperf3 -c host (client)	Throughput test	
iperf3 -c host -t 30 -i 5 -P 4			
smokeping	smokeping	Latency monitoring	
cacti	cacti	Network graphing	
munin	munin-node-configure	System monitoring	
zabbix	zabbix_agentd	Enterprise monitoring	
nagios	nagios	Network monitoring	

\$ NETWORKING - ATTACKS & DEFENSE

COMMON NETWORK ATTACKS

ATTACK	TOOL/COMMAND	DESCRIPTION	الدفاع
ARP Spoofing	arp spoof -i eth0 -t target gateway	Intercept traffic by fake ARP	arpwatch, static ARP
ettercap -T -M arp:remote // //			
DNS Spoofing	dnsspoof -i eth0 -f hosts.txt	Fake DNS responses	DNS Spoofing
ettercap -T -M dns:fake // //			
STP Spoofing	yersinia -G	Spanning Tree attack	BPDU Guard
DHCP Starvation	yersinia -I	Exhaust DHCP pool	DHCP Snooping
dhcpstarv eth0			
VLAN Hopping	yersinia -G	Hop between VLANs	disable VLAN
setportswitch enable			
SYN Flood	hping3 -S --flood -p 80 IP	TCP handshake DoS	SYN cookies
ICMP Flood	hping3 -I --flood IP	Ping flood	Rate limiting
UDP Flood	hping3 -2 --flood -p 53 IP	UDP packet flood	Rate limiting
MAC Flood	macof -i eth0	Flood switch CAM table	Port Security
MiTM	bettercap -eval 'net.sniff on; arp.spoof on'	Man-in-the-middle	HTTPS, mTLS, Certificate Pinning
mitmproxy -p 8080			
Replay Attack	tcpreplay -i eth0 capture.pcap	Replay captured packets	Timestamping

NETWORK HARDENING CHECKLIST

- CHECKLIST / قائمة التحقق
- [1] Change default passwords & credentials
 - [2] Disable unused services & ports (netstat -tulpn)
 - [3] Enable firewall: ufw enable or iptables -P INPUT DROP
 - [4] Use SSH keys, disable root login, change default port 22
 - [5] Enable SELinux or AppArmor (aa-status / getenforce)
 - [6] Segment network with VLANs and ACLs
 - [7] Enable port security on switches (port-security max 1)
 - [8] Use DHCP Snooping & Dynamic ARP Inspection
 - [9] Enable BPDU Guard & Root Guard (STP)
 - [10] Disable CDP/LLDP on untrusted ports
 - [11] Use VPN for remote access (WireGuard/OpenVPN)
 - [12] Implement IDS/IPS (Suricata/Snort)
 - [13] Enable logging & monitoring (auditd, rsyslog)
 - [14] Regular updates: apt update && apt upgrade
 - [15] Use encrypted protocols: HTTPS, SSH, SFTP instead of HTTP, Telnet, FTP
 - [16] Configure fail2ban for brute-force protection
 - [17] Implement 802.1X for network access control
 - [18] Use WPA2-Enterprise for Wi-Fi (not WPA2-PSK)
 - [19] Backup configurations regularly
 - [20] Conduct regular vulnerability scans & penetration tests

TROUBLESHOOTING METHODOLOGY

STEP	COMMAND	CHECK
1. Physical	ip link show eth0 grep -i state	Cable connected, link up
2. IP Config	ip addr show eth0 / ifconfig eth0	IP, subnet, gateway assigned
3. Gateway	ip route grep default	Default gateway exists
4. DNS	dig google.com / nslookup google.com	DNS resolution works
5. Local Ping	ping -c 4 gateway_IP	Can reach gateway
6. Remote Ping	ping -c 4 8.8.8.8	Internet connectivity
7. Port Check	nc -zv target port	Specific port open
8. Traceroute	traceroute target	Find where packets drop
9. Firewall	iptables -L -v -n / sudo ufw status	Rules blocking traffic
10. Listeners	ss -tulpn / netstat -tulpn	Service running
11. ARP	arp -a / ip neigh	MAC resolution
12. Logs	tail -f /var/log/syslog	System errors
13. Bandwidth	iperf3 -c server	Throughput test
14. Packet Capture	tcpdump -i eth0 host target	See actual packets

\$ NETWORKING - WI-FI & ADVANCED TOPICS

WI-FI STANDARDS & SECURITY

STANDARD	FREQUENCY	SPEED	SECURITY
802.11a	5 GHz	54 Mbps	WEP/WPA
802.11b	2.4 GHz	11 Mbps	WEP/WPA
802.11g	2.4 GHz	54 Mbps	WEP/WPA/WPA2
802.11n (Wi-Fi 4)	2.4/5 GHz	600 Mbps	WPA2
802.11ac (Wi-Fi 5)	5 GHz	3.5 Gbps	WPA2/WPA3
802.11ax (Wi-Fi 6)	2.4/5/6 GHz	9.6 Gbps	WPA3
802.11be (Wi-Fi 7)	2.4/5/6 GHz	46 Gbps	WPA3

WI-FI SECURITY TYPES / أنواع أمن الواي فاي

WEP (broken - cracked in minutes) | WPA-TKIP (deprecated) | WPA2-AES (current standard, PSK vulnerable to MITM) | WPA3 (strongest, protects against dictionary, forward secrecy) | WPA2-Enterprise (uses RADIUS, 802.1X, most organizations)

VLAN CONCEPTS

اضحية

VLAN / الشبكة المحلية الافتراضية

VLAN = Virtual LAN - logical segmentation of network at Layer 2. Isolates traffic without physical separation. 1-4095 (1 default, 2-1001 normal, 1002-4095 extended). Trunk ports carry multiple VLANs (802.1Q tagging). . single VLAN.

CONCEPT	DESCRIPTION	الوصف
Access Port	Single VLAN, for end devices	
Trunk Port	Multiple VLANs with 802.1Q tags	
Native VLAN	Untagged traffic on trunk (VLAN 1)	
VLAN Hopping	Switch Spoofing / Double Tagging	
VTP	VLAN Trunking Protocol (Cisco)	
PVLAN	Private VLAN - isolates ports within VLAN	
Voice VLAN	Separate VLAN for IP phones	
Management VLAN	VLAN for management traffic	

NETWORK TOPOLOGIES

TOPOLOGY	PROS	CONS	USA
Star	Easy management, fault isolation	Single point of failure (switch)	Mos
Bus	Simple, cheap	Single cable failure stops all	Leg
Ring	Fair access, predictable	Complex failure recovery	FDD
Mesh (Full)	Redundant, fault tolerant	Expensive, complex cabling	Bac
Mesh (Partial)	Cost effective redundancy	Complex routing	Lar
Tree	Scalable, hierarchical	Root failure affects all	Ent
Hybrid	Flexible, combines best of each	Complex management	Mos

